

Experiment#04

Objective:

To perform De Morgan's law practically with the help of AND/OR/NOT gates

Equipment's:

- Breadboard
- Jumper wires
- Power supply
- Integrated ICs
- LED light

Theory:

De Morgan's law:

De Morgan's law is the law that gives the relation between union, intersection and complements in set theory. In Boolean algebra, it gives the relation between AND, OR, and complements of the variable, and in logic, it gives the relation between AND, OR, or Negation of the statement. With the help of De Morgan's Law, we can optimize various boolean circuits involving logic gates which help us perform the same operation but with very few apparatus.

De Morgan's law in the set theory defines the relationship between the union, intersection, and complements of the sets, and is given for both complement of union and intersection of two sets. In set theory, there are two De Morgan's Laws that are:

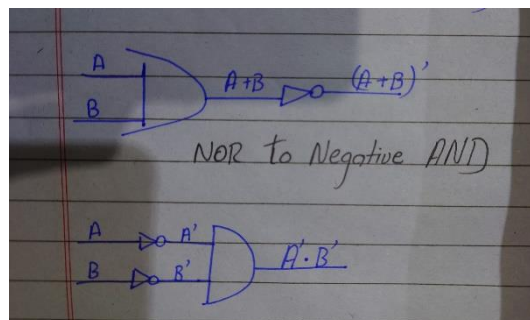
- First De Morgan's Law
- Second De Morgan's Law

First De Morgan's Law:

De Morgan's Law states that "Both the logic gate circuits i.e., NOT gate is added to the output of OR gate, and NOT gate is added to the input of AND gate, are equivalent.

Let A and B be two variables, then mathematically First De Morgan's Law is given as

$$(A+B)' = A' \cdot B'$$



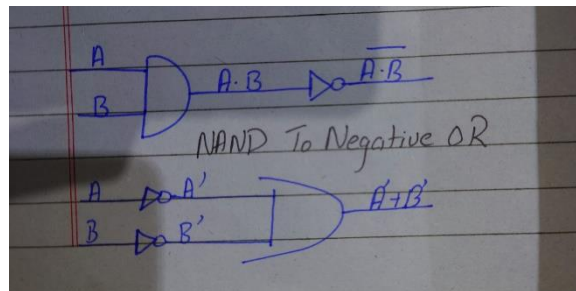
Fig#01

Second De Morgan's Law Logic Gates:

In context to logic gates and Boolean Algebra, De Morgan's Law states that "Both the logic gate circuits i.e., NOT gate is added to the output of AND gate, and NOT gate is added to the input of OR gate, are equivalent.

Let A and B be two variables, then mathematically Second De Morgan's Law is given as

$$(A.B)' = A' + B'$$

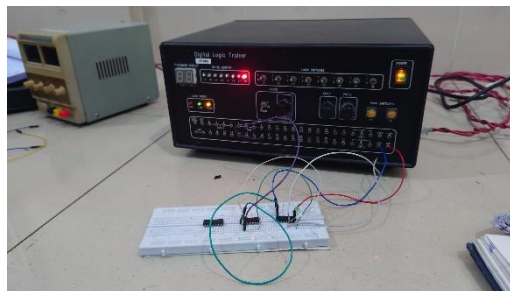


Fig#02

Procedure:

- First, collect all the required equipment and components that are used.
- Now make a circuit diagram according to the given equation.
- Make a circuit on the breadboard according to the circuit diagram.
- Put LED/DMM to check our circuit
- After making the circuit give a power supply to the circuit with the help of DC power supply and jumper wires.
- Check all the connections have been connected according circuit diagram.
- Now give supply to the circuit to make the truth table by changing values.

Circuit diagram:



Fig#03

Truth table:

First De Morgan's law

A	B	A+B	(A+B)'	A'	B'	A'.B'

Second De Morgan's law

A	B	A.B	(A.B)'	A'	B'	A'+B'

Conclusion:

In this experiment, we have learned how to make a circuit diagram with the help of equations and to make a circuit with the help of ICs on the breadboard. This experiment also helps to learn how NAND to negative OR and NOR to negative NAND gates are equal with the help of De Morgan's law.